

CAPSTONE PROJECT -02

JOB MARKET ANALYSIS

TEAM ID: PTID-CDA-OCT-25-803

PROJECT ID: PRDA-04 JOB MARKET ANALYSIS

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INTRODUCTION

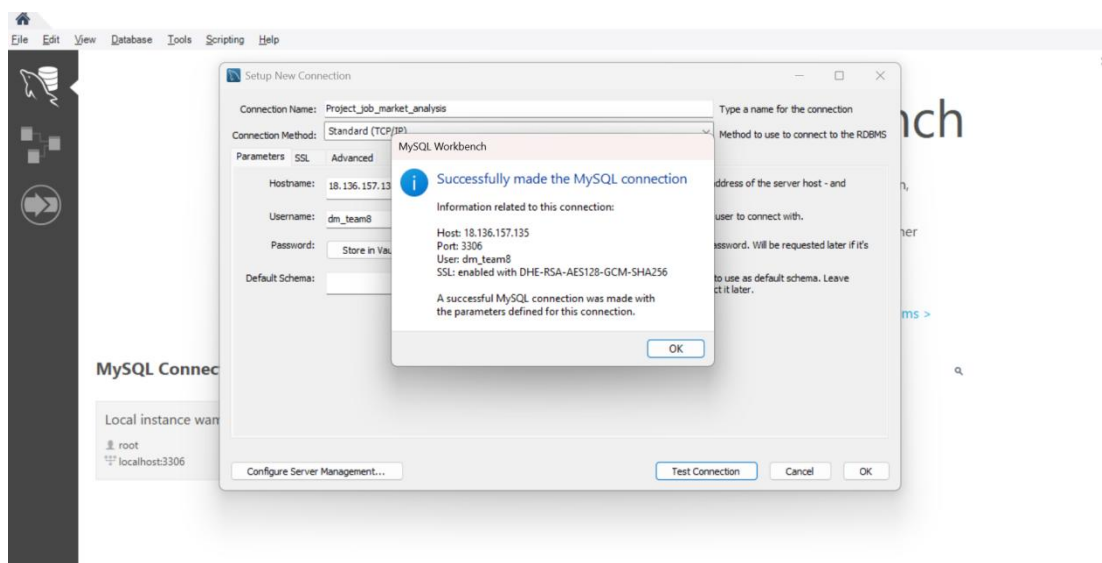
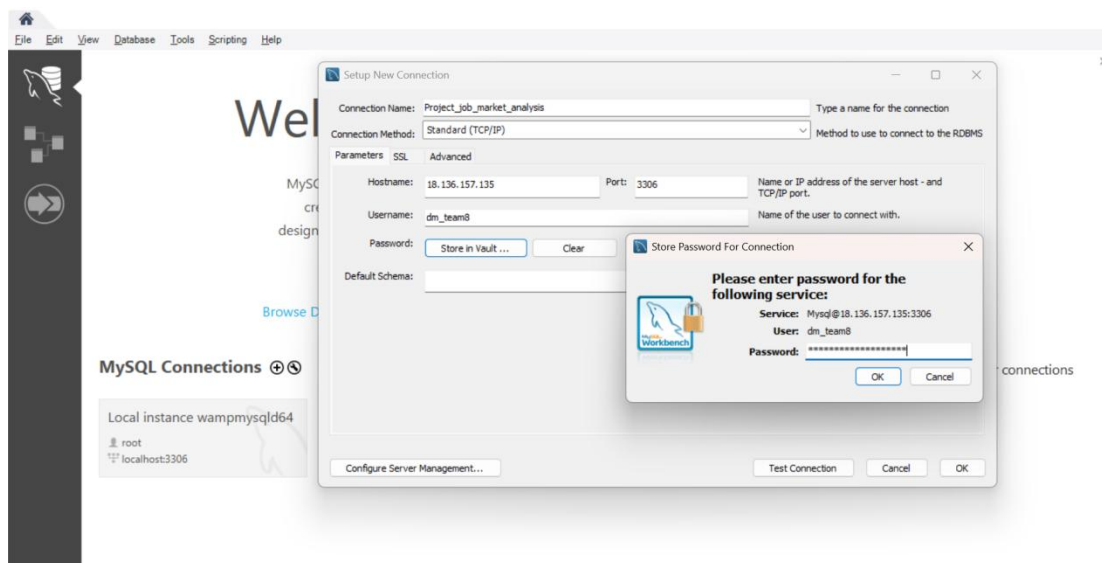
- In the modern data-driven economy, understanding job market trends is crucial for both job seekers and employers. The dataset consists of 742 records and 42 attributes, providing a comprehensive overview of job titles, salary ranges, company details, required skills, and other organizational factors. By examining these features, the study aims to identify which locations, industries, and qualifications are most in demand, helping stakeholders make informed career and business decisions.
- The main factors analyzed are: states with the highest number of job opportunities, average and range of salaries, and industries leading in Data Science job postings. Additionally, we will discuss trends associated with education levels and salary and the most frequent requirements by employers, too.
- Visualization and interpretation of data are effectively done using Tableau and Power BI. These visualizations uncover patterns and correlations that are not immediately apparent from the raw data. The insights drawn from this project will aid job seekers in identifying the most valuable skills and areas that pay the most, while companies benefit by understanding the competition in the market, salary benchmarks, and demand in the industry. The ultimate goal of this analysis is to bridge the gap in supply and demand in the workforce with more informed decisions in hiring, career planning, and skill development.

DATA ANALYSIS USING SQL

Q.1. Get data from the database with the given credentials.

We connect to the database using valid credentials (username, password, host) through a database driver. Once authenticated, we run SQL queries to retrieve the required data.

Screenshot:



Project_job_market_analysis x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

project_job_market_analysis

Tables

Market

Columns

- ID
- Job_Title
- Salary_Estimate
- FIELD4
- Rating
- Company_Name
- Location
- Headquarters
- Size
- Founded
- Type_of_ownership
- Industry

Administration

Schemas

Information

Table: Market

Columns:

- ID int(11) PK
- Job_Title varchar(98)
- Salary_Estimate varchar(41)
- FIELD4 varchar(30)
- Rating decimal(3,1)
- Company_Name varchar(55)
- Location varchar(33)

Query 1 Market

Limit to 1000 rows

1 • SELECT * FROM project_job_market_analysis.Market;

Result Grid

ID	Job_Title	Salary_Estimate	FIELD4	Rating	Company_Name	Location	Headquarters	Size
0	Data Scientist	\$53K-\$91K (Glassdoor est.)	NULL	3.8	Tecolote Research 3.8	Albuquerque, NM	Goleta, CA	501 - 1000
1	Healthcare Data Scientist	\$63K-\$112K (Glassdoor est.)	NULL	3.4	University of Maryland Medical Syste...	Linthicum, MD	Baltimore, MD	10000+
2	Data Scientist	\$80K-\$90K (Glassdoor est.)	NULL	4.8	KnowBe4 4.8	Clearwater, FL	Clearwater, FL	501 - 1000
3	Data Scientist	\$59K-\$97K (Glassdoor est.)	NULL	3.8	PHM 3.8	Richland, WA	Richland, WA	1001 - 5000
4	Data Scientist	\$86K-\$143K (Glassdoor est.)	NULL	2.9	Affinity Solutions 2.9	New York, NY	New York, NY	51 - 200
5	Data Scientist	\$71K-\$119K (Glassdoor est.)	NULL	3.4	CyrusOne 3.4	Dallas, TX	Dallas, TX	201 - 500
6	Data Scientist	\$54K-\$93K (Glassdoor est.)	NULL	4.1	ClearOne Advantage 4.1	Baltimore, MD	Baltimore, MD	501 - 1000
7	Data Scientist	\$86K-\$142K (Glassdoor est.)	NULL	3.8	Logic20/20 3.8	San Jose, CA	Seattle, WA	201 - 500
8	Research Scientist	\$38K-\$84K (Glassdoor est.)	NULL	3.3	Rochester Regional Health 3.3	Rochester, NY	Rochester, NY	10000+
9	Data Scientist	\$120K-\$160K (Glassdoor est.)	NULL	4.6	<content> 4.6	New York, NY	New York, NY	51 - 200
10	Data Scientist	\$126K-\$201K (Glassdoor est.)	NULL	3.5	Wish 3.5	San Jose, CA	San Francisco...	501 - 1000
11	Data Scientist	\$64K-\$106K (Glassdoor est.)	NULL	4.1	ManTech 4.1	Chantilly, VA	Herndon, VA	5001 - 10000
12	Staff Data Scientist - Te...	\$106K-\$172K (Glassdoor est.)	NULL	3.2	Walmart 3.2	Plano, TX	Bentonville, AR	10000+

Market 1 x

Apply Revert

Q.2.States with most number of Jobs.

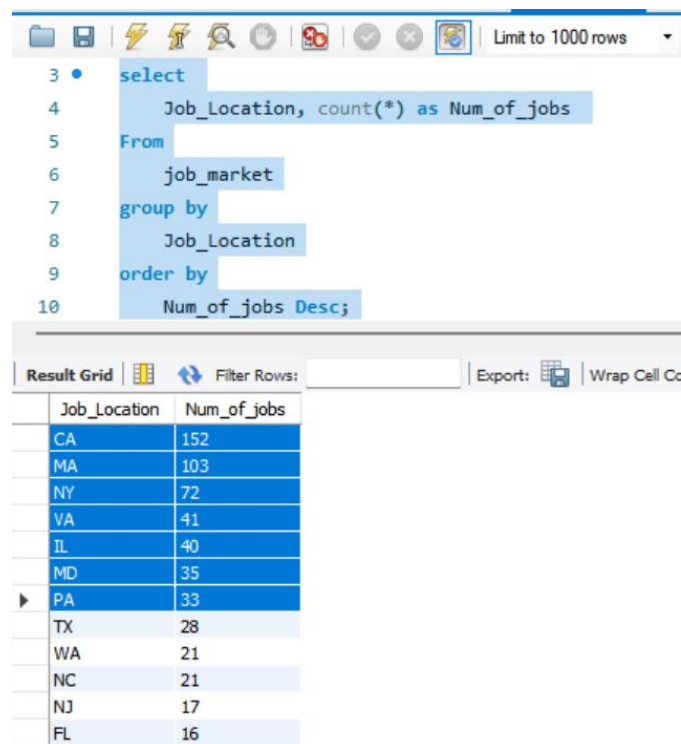
Query:

```
select Job_Location, count (*) as Num_of_jobs
```

```
From job_market
```

```
group by Job_Location
```

```
order by Num_of_jobs Desc;
```



Limit to 1000 rows

```
3 • select
4     Job_Location, count(*) as Num_of_jobs
5 From
6     job_market
7 group by
8     Job_Location
9 order by
10    Num_of_jobs Desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Cc

Job_Location	Num_of_jobs
CA	152
MA	103
NY	72
VA	41
IL	40
MD	35
PA	33
TX	28
WA	21
NC	21
NJ	17
FL	16

Q.3. Average Minimal and Maximal Salaries in Different States.

Query:

```
SELECT
```

```
    Job_Location,
```

```
    MIN(Avg_Salary_in_k) AS Average_Minimal_Salary,
```

```
    MAX(Avg_Salary_in_k) AS Average_Maximal_Salary
```

```
FROM
```

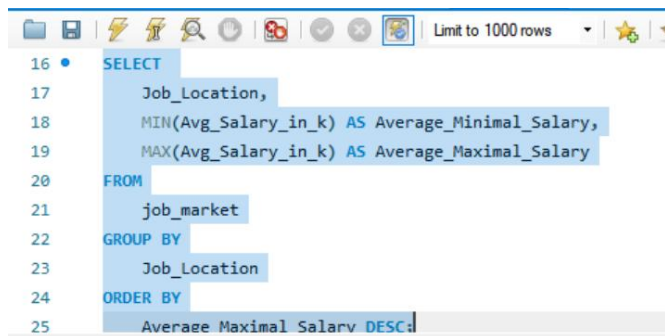
job_market

GROUP BY

Job_Location

ORDER BY

Average_Maximal_Salary DESC;



```
16 • SELECT
17     Job_Location,
18     MIN(Avg_Salary_in_k) AS Average_Minimal_Salary,
19     MAX(Avg_Salary_in_k) AS Average_Maximal_Salary
20 FROM
21     job_market
22 GROUP BY
23     Job_Location
24 ORDER BY
25     Average_Maximal_Salary DESC;
```

Result Grid

Job_Location	Average_Minimal_Salary	Average_Maximal_Salary
IL	52.5	254
DC	50	237.5
CA	47.5	232.5
WA	44.5	184.5
NJ	56.5	173
NY	15.5	172
MA	47.5	172
PA	42	155
TX	44.5	149.5

Q.4. Average Salary in Different States.

Query:

Select Job_Location,

Round (avg ((Lower_Salary_in_Thousand_\$ + Upper_Salary_in_Thousand_\$)/2),2)
as Average_Salary_in_K_\$

from job_market

group by Job_Location

order by Average_Salary_in_K_\$ Desc;

Screenshot:

```
33 • Select
34     Job_Location,
35     Round(avg((Lower_Salary_in_Thousand_$ + Upper_Salary_in_Thousand_$)/2),2) as Average_Salary_in_K_$
36 from
37     job_market
38 group by
39     Job_Location
40 order by
41     Average_Salary_in_K_$ Desc;
```

Result Grid

Job_Location	Average_Salary_in_K_\$
CA	123.51
IL	116.66
DC	110.18
MA	107.50
NJ	104.56
MI	100.25
RI	100.00
NY	98.65
NC	98.45

Result 19 x

Q.5. Top 5 Industries with maximum number of Data Science related job postings.

Query:

```
SELECT Industry,
COUNT (*) AS Job_post
FROM job_market
WHERE Job_Title REGEXP 'Scientist|Science'
GROUP BY
Industry
ORDER BY Job_post DESC LIMIT 5;
```

```
43 • SELECT
44     Industry,
45     COUNT(*) AS Job_post
46 FROM
47     job_market
48 WHERE
49     Job_Title REGEXP 'Scientist|Science'
50 GROUP BY
51     Industry
52 ORDER BY
53     Job_post DESC
54 LIMIT 5;
```

Result Grid

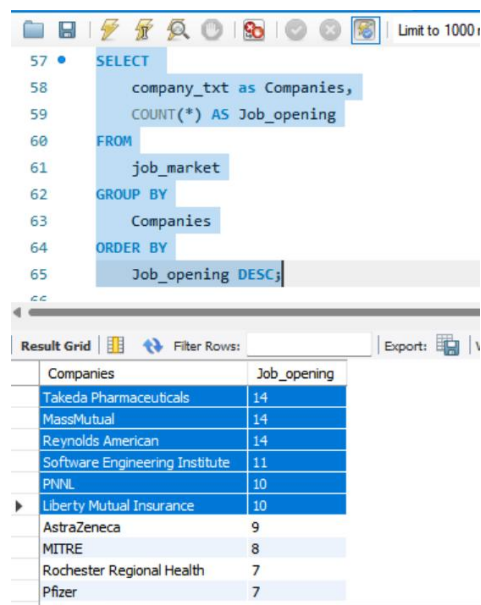
Industry	Job_post
Biotech & Pharmaceuticals	104
Health Care Services & Hospitals	32
Computer Hardware & Software	30
Insurance Carriers	28
Enterprise Software & Network Solutions	25

Result 26 x

Q.6. Companies with maximum Number of job openings.

Query:

```
SELECT company_txt as Companies,  
COUNT (*) AS Job_opening  
FROM job_market  
GROUP BY  
Companies ORDER BY Job_opening DESC;
```



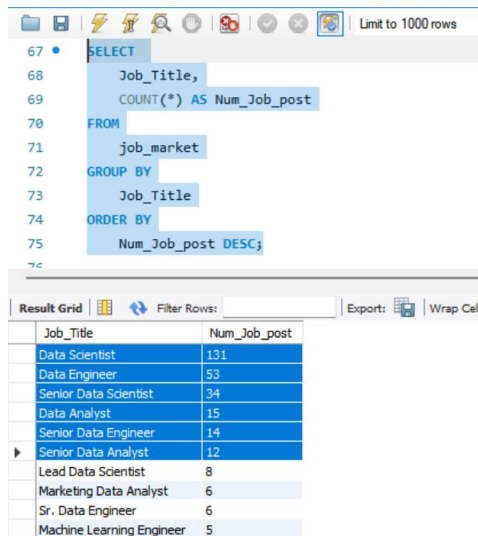
Result Grid

Companies	Job_opening
Takeda Pharmaceuticals	14
MassMutual	14
Reynolds American	14
Software Engineering Institute	11
PNINL	10
Liberty Mutual Insurance	10
AstraZeneca	9
MITRE	8
Rochester Regional Health	7
Pfizer	7

Q.7. Job Titles with most Number of Jobs.

Query:

```
SELECT company_txt as Companies,  
COUNT (*) AS Job_opening  
FROM job_market  
GROUP BY Companies  
ORDER BY Job_opening DESC;
```



```

67 • SELECT
68     Job_Title,
69     COUNT(*) AS Num_Job_post
70 FROM
71     job_market
72 GROUP BY
73     Job_Title
74 ORDER BY
75     Num_Job_post DESC;
76

```

Job_Title	Num_Job_post
Data Scientist	131
Data Engineer	53
Senior Data Scientist	34
Data Analyst	15
Senior Data Engineer	14
Senior Data Analyst	12
Lead Data Scientist	8
Marketing Data Analyst	6
Sr. Data Engineer	6
Machine Learning Engineer	5

Q.8. Salary of job titles with most number of jobs.

Query:

SELECT Job_Title,

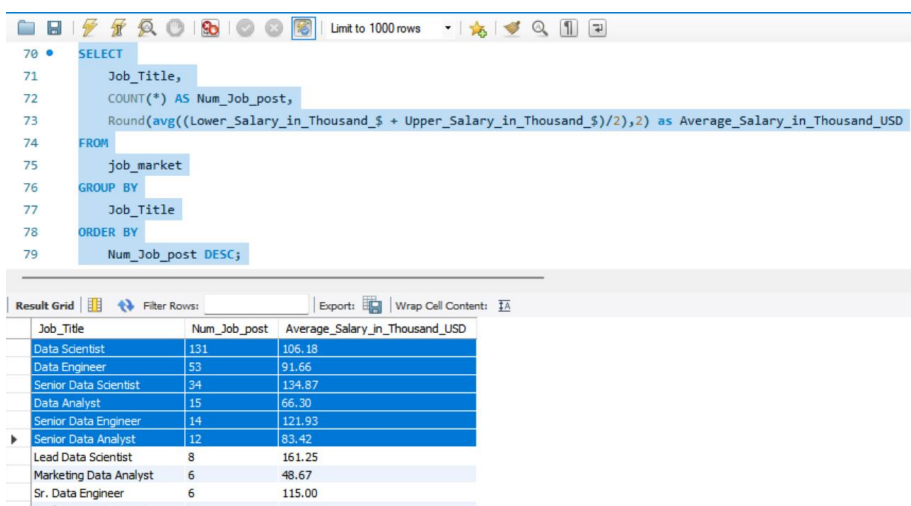
COUNT (*) AS Num_Job_post,

Round (avg ((Lower_Salary_in_Thousand_\$ + Upper_Salary_in_Thousand_\$)/2),2)
as Average_Salary_in_Thousand_USD

FROM job_market

GROUP BY Job_Title

ORDER BY Num_Job_post DESC, Average_Salary_in_Thousand_USD DESC;



```

70 • SELECT
71     Job_Title,
72     COUNT(*) AS Num_Job_post,
73     Round(avg((Lower_Salary_in_Thousand_$ + Upper_Salary_in_Thousand_$)/2),2) as Average_Salary_in_Thousand_USD
74 FROM
75     job_market
76 GROUP BY
77     Job_Title
78 ORDER BY
79     Num_Job_post DESC;

```

Job_Title	Num_Job_post	Average_Salary_in_Thousand_USD
Data Scientist	131	106.18
Data Engineer	53	91.66
Senior Data Scientist	34	134.87
Data Analyst	15	66.30
Senior Data Engineer	14	121.93
Senior Data Analyst	12	83.42
Lead Data Scientist	8	161.25
Marketing Data Analyst	6	48.67
Sr. Data Engineer	6	115.00

Q.9. Skills required by companies for each job title.

Query:

```
SELECT
  Company_txt as Companies,
  Job_Title,
  TRIM (BOTH ',' FROM CONCAT_WS (',',
    IF (python = 1, 'Python', NULL),
    IF (spark = 1, 'Spark', NULL),
    IF (aws = 1, 'AWS', NULL),
    IF (excel = 1, 'Excel', NULL),
    IF (sas = 1, 'SAS', NULL),
    IF (keras = 1, 'Keras', NULL),
    IF (pytorch = 1, 'PyTorch', NULL),
    IF (scikit = 1, 'Scikit-Learn', NULL),
    IF (tensor = 1, 'TensorFlow', NULL),
    IF (hadoop = 1, 'Hadoop', NULL),
    IF (tableau = 1, 'Tableau', NULL),
    IF (bi = 1, 'Power BI', NULL),
    IF (flink = 1, 'Flink', NULL),
    IF (mongo = 1, 'MongoDB', NULL),
    IF (google_an = 1, 'Google Analytics', NULL)
  )) AS Skills_Required
FROM job_market
ORDER BY Company_txt, Job_Title;
```

```

84 • SELECT
85     Company_txt as Companies,
86     Job_Title,
87     TRIM(BOTH ' ' FROM CONCAT_WS(' ',
88         IF(python = 1, 'Python', NULL),
89         IF(spark = 1, 'Spark', NULL)

```

Companies	Job_Title	Skills_Required
Acuity Insurance	Senior Research Statistician- Data Scientist	SAS
Adobe	Data Science Engineer - Mobile	Python,Excel,Hadoop,Tableau,Power BI
Adobe	Data Science Engineer - Mobile	Python,Excel,Hadoop,Tableau,Power BI
Advanced BioScience Labo...	Quality Control Scientist III- Analytical Develop...	Excel
Advanced BioScience Labo...	Staff Scientist- Upstream PD	
Advanced BioScience Labo...	Staff Scientist- Upstream PD	
Advanced BioScience Labo...	Staff Scientist-Downstream Process Development	Excel
Advanced BioScience Labo...	Staff Scientist-Downstream Process Development	Excel
Advanced BioScience Labo...	Staff Scientist-Downstream Process Development	Excel
Advanced BioScience Labo...	Staff Scientist-Downstream Process Development	Excel
Affinity Solutions	Data Scientist	Python,Excel,SAS
Affinity Solutions	Staff BI and Data Engineer	Python,Spark,Tableau
Affinity Solutions	Staff BI and Data Engineer	Python,Spark,Tableau
Agios Pharmaceuticals	Manager, Safety Scientist, Medical Safety & Ris...	Excel
Agios Pharmaceuticals	Scientist, Pharmacometrics	SAS

Q.10. Analyse all the features and derive multiple insights.

1. Cities with most openings.

Query:

```

SELECT * FROM jma.job_market;
SELECT Job_City, Job_State, COUNT (*) AS postings
FROM job_market
GROUP BY Job_City, Job_State
ORDER BY postings DESC
LIMIT 20;

```

```

143 • SELECT * FROM jma.job_market;
144 • SELECT Job_City, Job_State, COUNT(*) AS postings
145 FROM job_market
146 GROUP BY Job_City, Job_State
147 ORDER BY postings DESC
148 LIMIT 20;
149 -----
150 • SELECT * FROM jma.job_market;
151

```

Job_City	Job_State	postings
New York	NY	55
San Francisco	CA	49
Cambridge	MA	47
Chicago	IL	32
Boston	MA	23
San Jose	CA	13
Pittsburgh	PA	12
Washington	DC	11
Rockville	MD	11
Richland	WA	10
Hamilton	VA	10

Result 17 x

2. “Data Science” related vs others (volume & salary)

Query:

SELECT CASE

WHEN Job_Title REGEXP 'Scientist|Science' THEN 'Data Science related'

ELSE 'Other'

END AS Science_bucket,

COUNT (*) AS postings,

Round (avg ((Lower_Salary_in_Thousand_\$ +
Upper_Salary_in_Thousand_\$)/2),2) as Average_Salary_in_K_\$

FROM job_market

GROUP BY Science_bucket

ORDER BY postings DESC;

```

152 • SELECT
153 CASE
154 WHEN Job_Title REGEXP 'Scientist|Science' THEN 'Data Science related'
155 ELSE 'Other'
156 END AS Science_bucket,
157 COUNT(*) AS postings,
158 Round(avg((Lower_Salary_in_Thousand_$ + Upper_Salary_in_Thousand_$)/2),2) as Average_Salary_in_K_$
159 FROM job_market
160 GROUP BY Science_bucket
161 ORDER BY postings DESC;

```

Science_bucket	postings	Average_Salary_in_K_\$
Data Science related	472	109.54
Other	270	87.40

Result 16 x

Output

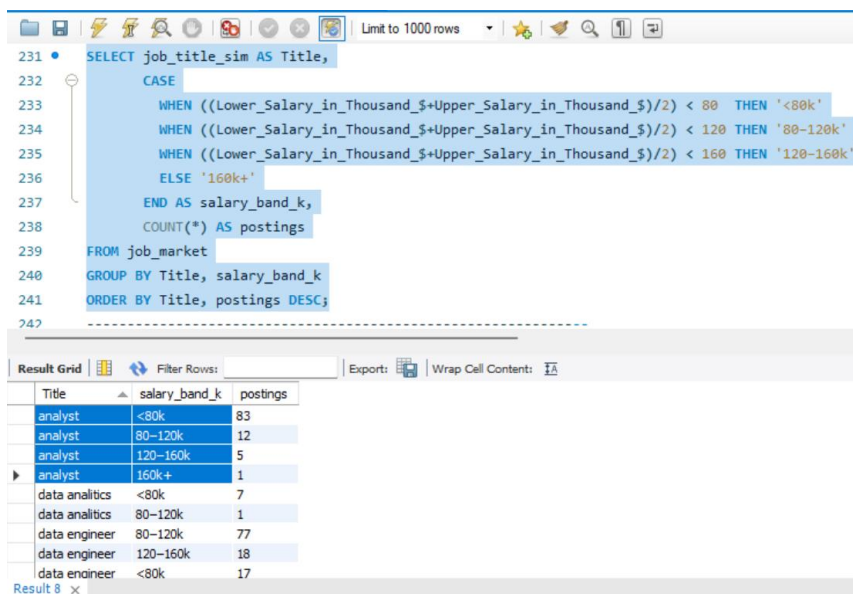
Action Output

#	Time	Action	Message
28	11:20:25	SELECT CASE WHEN Job_Title REGEXP 'Scientist Science' THEN 'Da...	Error Code: 1055. Expression #3 of SELECT list is not in GROUP BY
29	11:22:49	SELECT CASE WHEN Job_Title REGEXP 'Scientist Science' THEN 'Da...	2 row(s) returned

3. Salary bands per title (distribution)

Query:

```
SELECT job_title_sim AS Title,  
CASE WHEN  
((Lower_Salary_in_Thousand_$$Upper_Salary_in_Thousand_$)/2) < 80  
THEN '<80k'  
WHEN ((Lower_Salary_in_Thousand_$$Upper_Salary_in_Thousand_$)/2) <  
120 THEN '80-120k'  
WHEN ((Lower_Salary_in_Thousand_$$Upper_Salary_in_Thousand_$)/2) <  
160 THEN '120-160k'  
ELSE '160k+'  
END AS salary_band_k,  
COUNT (*) AS postings  
FROM job_market  
GROUP BY Title, salary_band_k  
ORDER BY Title, postings DESC;
```



```
231 SELECT job_title_sim AS Title,  
232 CASE  
233 WHEN ((Lower_Salary_in_Thousand_$$Upper_Salary_in_Thousand_$)/2) < 80 THEN '<80k'  
234 WHEN ((Lower_Salary_in_Thousand_$$Upper_Salary_in_Thousand_$)/2) < 120 THEN '80-120k'  
235 WHEN ((Lower_Salary_in_Thousand_$$Upper_Salary_in_Thousand_$)/2) < 160 THEN '120-160k'  
236 ELSE '160k+'  
237 END AS salary_band_k,  
238 COUNT(*) AS postings  
239 FROM job_market  
240 GROUP BY Title, salary_band_k  
241 ORDER BY Title, postings DESC;
```

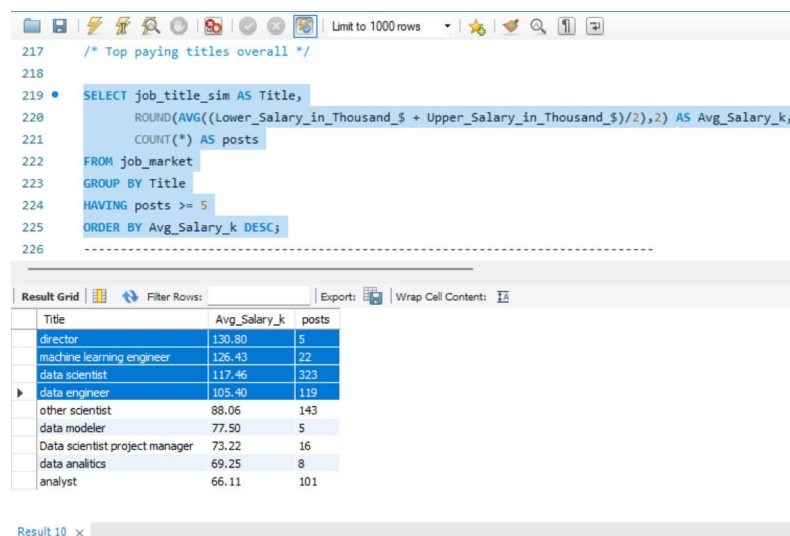
Title	salary_band_k	postings
analyst	<80k	83
analyst	80-120k	12
analyst	120-160k	5
analyst	160k+	1
data analytics	<80k	7
data analytics	80-120k	1
data engineer	80-120k	77
data engineer	120-160k	18
data engineer	<80k	17

Result 8 x

4. Top paying titles overall

Query:

```
SELECT job_title_sim AS Title,  
ROUND (AVG ((Lower_Salary_in_Thousand_$ +  
Upper_Salary_in_Thousand_$)/2),2) AS Avg_Salary_k,  
COUNT (*) AS posts  
FROM job_market  
GROUP BY Title  
HAVING posts >= 5  
ORDER BY Avg_Salary_k DESC;
```



Title	Avg_Salary_k	posts
director	130.80	5
machine learning engineer	126.43	22
data scientist	117.46	323
data engineer	105.40	119
other scientist	88.06	143
data modeler	77.50	5
Data scientist project manager	73.22	16
data analytics	69.25	8
analyst	66.11	101

5. Most in-demand skills (overall)

Query:

```
SELECT skill, total_postings  
FROM (  
SELECT 'python' AS skill, SUM (python) AS total_postings FROM job_market  
UNION ALL  
SELECT 'SQL', SUM (SQL) FROM job_market UNION ALL  
SELECT 'aws', SUM (aws) FROM job_market UNION ALL  
SELECT 'spark', SUM (spark) FROM job_market UNION ALL  
SELECT 'hadoop', SUM (hadoop) FROM job_market UNION ALL  
SELECT 'tableau', SUM (tableau) FROM job_market UNION ALL  
SELECT 'excel', SUM (excel) FROM job_market) s  
ORDER BY total_postings DESC;
```

Query 1 actor hr_da Job_market_DA x job_market

Limit to 1000 rows

```

195
196 • SELECT skill, total_postings
197 FROM (
198     SELECT 'python' AS skill, SUM(python) AS total_postings FROM job_market UNION ALL
199     SELECT 'sql_', SUM(sql_) FROM job_market UNION ALL
200     SELECT 'aws', SUM(aws) FROM job_market UNION ALL
201     SELECT 'spark', SUM(spark) FROM job_market UNION ALL
202     SELECT 'hadoop', SUM(hadoop) FROM job_market UNION ALL
203     SELECT 'tableau', SUM(tableau) FROM job_market UNION ALL
204     SELECT 'excel', SUM(excel) FROM job_market
205 ) s
206 ORDER BY total_postings DESC;
207

```

Result Grid

skill	total_postings
python	392
excel	388
sql_	380
aws	176
spark	167
tableau	148
hadoop	124

Result 12 x

6. Salary spread within each job role

Query:

```

SELECT job_title_sim AS Title,
COUNT (*) AS postings,
MIN(`Lower_Salary_in_Thousand_$`) AS min_salary_k,
MAX(`Upper_Salary_in_Thousand_$`) AS max_salary_k,
ROUND (AVG ((`Lower_Salary_in_Thousand_$` + `Upper_Salary_in_Thousand_$`)
/ 2), 2) AS avg_salary_k
FROM job_market
WHERE `Lower_Salary_in_Thousand_$` IS NOT NULL
AND `Upper_Salary_in_Thousand_$` IS NOT NULL
GROUP BY Title
ORDER BY postings DESC;

```


281	/*Salary spread within each Job role*/
282	SELECT
283	job_title_sim AS Title,
284	COUNT(*) AS postings,
285	MIN(`Lower_Salary_in_Thousand_\$`) AS min_salary_k,
286	MAX(`Upper_Salary_in_Thousand_\$`) AS max_salary_k,
287	ROUND(AVG((`Lower_Salary_in_Thousand_\$` + `Upper_Salary_in_Thousand_\$`) / 2), 2) AS avg_salary_k
288	FROM job_market
289	WHERE `Lower_Salary_in_Thousand_\$` IS NOT NULL
290	AND `Upper_Salary_in_Thousand_\$` IS NOT NULL
291	GROUP BY Title
292	ORDER BY postings DESC;
293	

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Title	postings	min_salary_k	max_salary_k
data scientist	323	15	306
other scientist	143	29	231
data engineer	119	42	228
analyst	101	20	228
machine learning engineer	22	61	289
Data scientist project manager	16	26	134
data analytics	8	37	138

Result 1 x

7. Average salary by seniority (within each title).

Query:

```
SELECT
  job_title_sim AS Title,
  seniority_by_title AS Seniority,
  ROUND (AVG ((`Lower_Salary_in_Thousand_$` +
`Upper_Salary_in_Thousand_$`) / 2), 2) AS Avg_Salary_k,
  COUNT (*) AS posts
FROM job_market
GROUP BY Title, Seniority
ORDER BY Title, Avg_Salary_k DESC;
```


CHALLENGES DURING EXECUTION

Challenges and Solutions

1. Challenge: Dataset contained inconsistent, missing, and incorrectly formatted values.

Solution: Performed detailed data cleaning using Excel/SQL functions to remove nulls, fix formats, and ensure consistency before querying.

2. Challenge: Large dataset size and many columns resulted in slow query execution and difficulty identifying relevant fields.

Solution: Applied indexing, optimized SQL queries, and selected only necessary columns to improve performance and readability.

3. Challenge: Unclear or inconsistent column names made interpretation difficult.

Solution: Referred to dataset documentation/metadata and used SQL aliasing or renaming to make columns meaningful and easier to understand.

4. Challenge: Incorrect or inconsistent data types (e.g., numbers stored as text) caused errors during analysis.

Solution: Converted data types using SQL functions to ensure accurate filtering, aggregation, and calculations.

5. Challenge: Duplicate records across multiple columns and tables affected accuracy.

Solution: Identified and removed duplicate entries to eliminate redundancy and maintain data reliability.

6. Challenge: Long text fields and too many columns made visualization and chart selection difficult.

Solution: Simplified data by selecting essential fields, summarizing text columns, and using sample data to create clear and effective visualizations.

7. Challenge: Selecting suitable and visually balanced colour combinations for charts was difficult.

Solution: Experimented with different colour palettes and used Power BI's theme customization feature to ensure consistent and professional visuals.

8. Challenge: Crafting meaningful storytelling, clear titles, and impactful insights was challenging.

Solution: Analysed results carefully, identified key trends, and used concise, insight-driven language to improve storytelling quality.

BUSINESS INSIGHTS

- Data driven roles dominate hiring, with Data Scientist and Data Engineer leading, showing companies' strong push toward analytics and automation.
- Python, SQL, and Excel remain the most essential skills, that means the foundational technical capabilities are crucial in most jobs related to data.
- Junior positions are in highest demand, meaning organizations prefer talent that will scale, be cost-efficient, and grow with them.
- Increased average salary for Master's Degree holders. This shows that advanced education provides massive returns related to career and compensation.
- Industries like Biotech, Pharma, Insurance, and IT show the highest job volume, signaling where market growth and digital transformation are strongest.
- States like CA, IL, and DC offer the best opportunities in terms of salary and indicate geographic clusters where companies are willing to pay a premium for skilled talent.

POWERBI DASHBOARDS

INTRODUCTON

This report provides a comprehensive analysis of the U.S. job market, focusing on state-wise trends, industry demand, and the relationship between skills, education, and salary expectations. The insights are organized into three analytical patterns designed to highlight where opportunities exist, which roles and industries are growing, and what qualifications employers value most. The objective is to support job seekers, workforce planners, and policymakers in making informed, data-driven decisions.

Pattern-1: State Job Trends

Examines how job availability and salary levels vary across states. This pattern reveals high-demand locations, competitive salary regions, and geographic clusters of job opportunities helping users identify where specific roles are most rewarding and accessible.

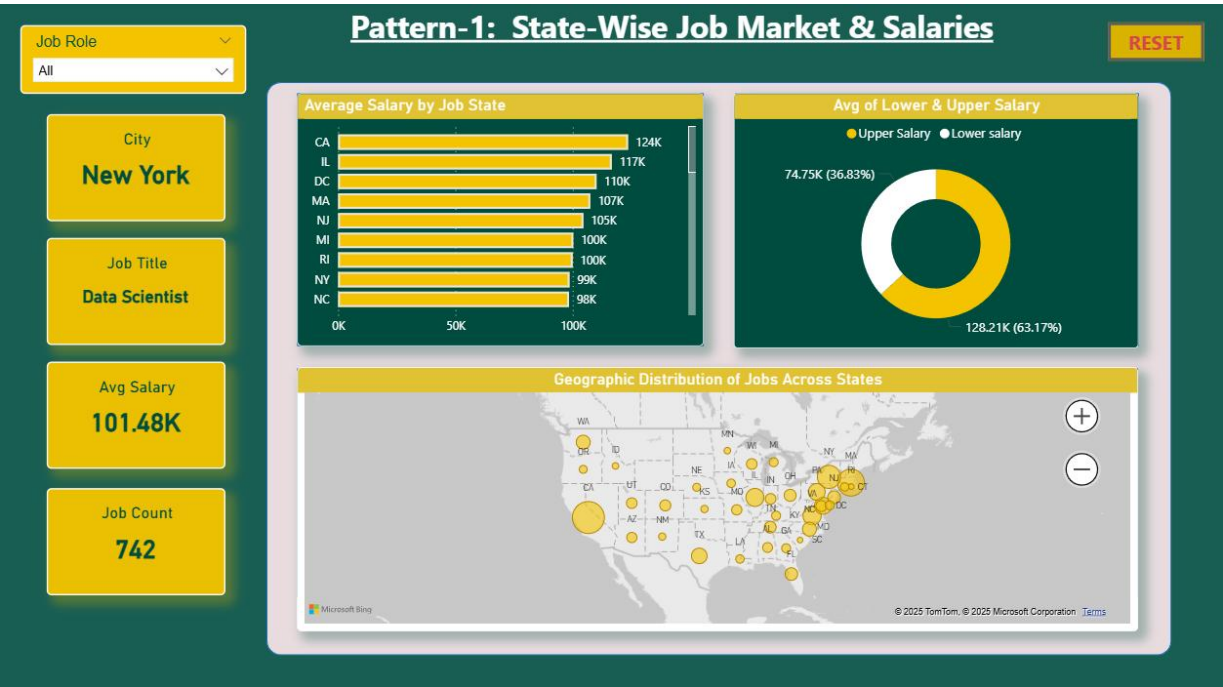
Pattern-2: Industry Job Insights

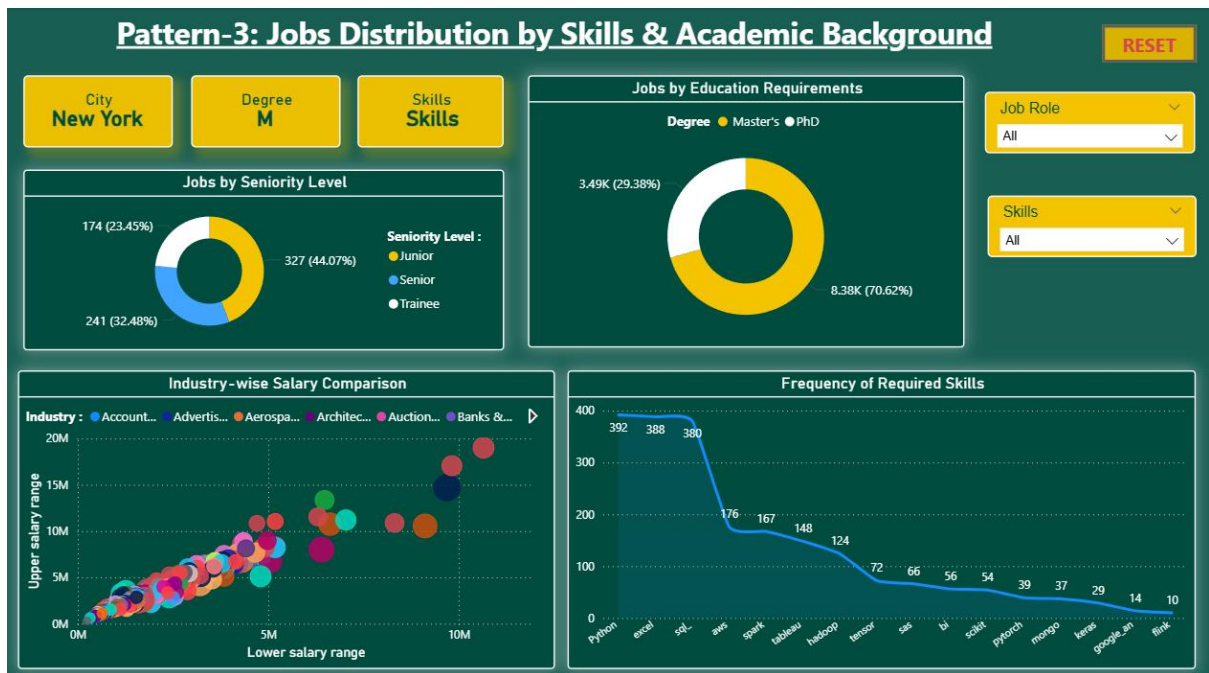
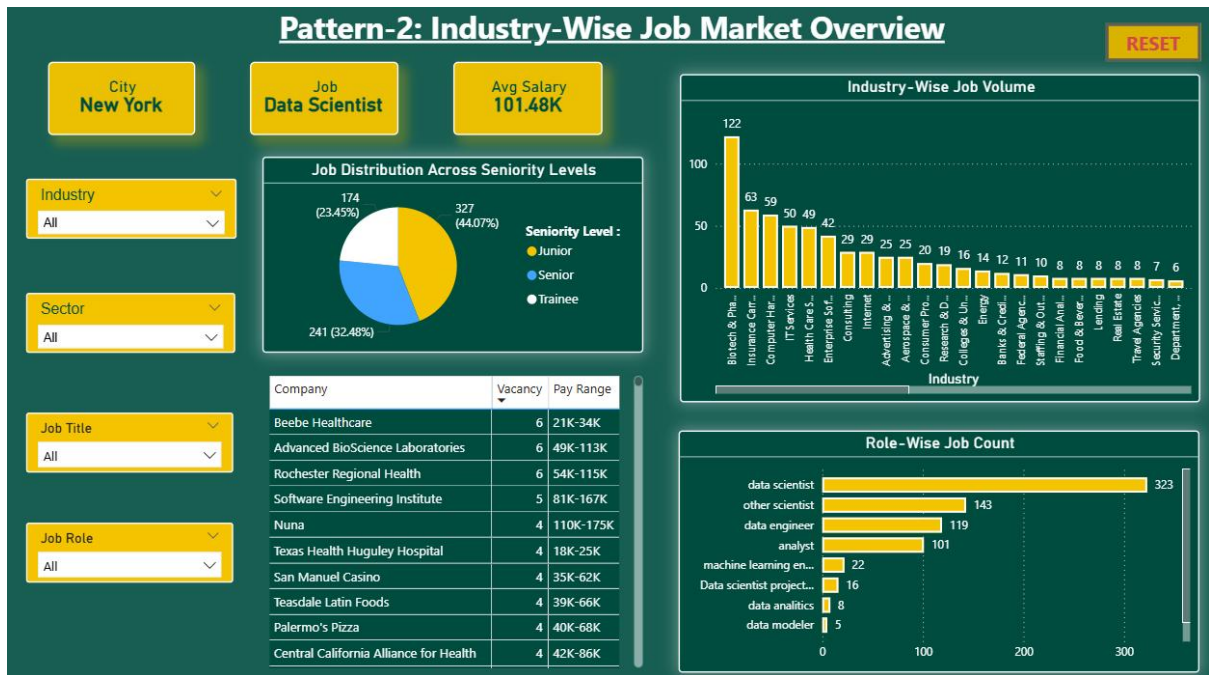
Analyzes job demand across industries, sectors, and seniority levels. This pattern uncovers which industries are hiring the most, what roles dominate the job market, and how vacancy counts differ across companies supporting strategic career planning and industry selection.

Pattern-3: Skills & Education Impact

Explores how required skills and educational qualifications influence job availability and salary ranges. This pattern highlights the most in-demand technical skills, the value of advanced degrees, and salary variations across industries guiding learners and professionals on which capabilities to develop for career advancement.

Note: All project questions and scenarios are covered within these patterns.





CONCLUSION

- Data Scientist, Data Engineer, and Analyst are leading the job market with a high demand for data-oriented talent. This further points out the continued industry movement toward analytics-driven decision-making.
- Python, Excel, and SQL are the most critical skills, these three form the foundation of most high demand roles. The investment in these core skills greatly enhances job readiness and competitiveness.
- Junior level positions dominate the hiring landscape, meaning firms prefer entry-level talent that is scalable. This trend is indicative of strong opportunities for freshers and early-career professionals.
- Master's degree holders have higher average salaries than PhDs, which is indicative of strong ROI from professional degrees. Education still plays a key role in compensation and career advancement.
- High paying states like CA, IL, and DC and growth heavy industries like Biotech, Pharma, and IT are the drivers of the job market. The clusters showcase the areas where job seekers can target to get better salaries and long-term growth.